Process Characterization Report

A-Mab Drug Substance — Small-Virus Retentive Filtration (Step 9) — PCR-009

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

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Approvals

Role	Function	Name (synthetic)	Signature	Date
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Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Synthetic record. Signatures are not applied.

Revision history

Version	Effective date	Description of change
1.0	2026-07-24	Initial issue
2.0	2026-09-30	Reissued under INV-001 and INV-002: a priori intermediate acceptance criteria, re-analysis of the studies, execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
ADCC	antibody-dependent cell-mediated cytotoxicity
AICc	Akaike’s information criterion with the small-sample correction
AMD	analytical method decision
AMV	analytical method validation
CI	confidence interval
Cpk	process capability index
CQA	critical quality attribute
CV	coefficient of variation
DoE	design of experiments
DS	drug substance
GMP	good manufacturing practice
HMW	high molecular weight species
IAC	intermediate acceptance criterion
ICH	International Council for Harmonisation
LMW	low molecular weight species
LOQ	limit of quantitation
LRF	log reduction factor
MSAT	Manufacturing Science and Technology
MVM	minute virus of mice
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
PAR	proven acceptable range
psi	pound-force per square inch
QA	Quality Assurance
qPCR	quantitative polymerase chain reaction
SD	standard deviation
SDM	scale-down model
SOP	standard operating procedure
TCID50	median tissue culture infective dose

Abbreviation	Meaning
WC-CPP	well-controlled critical process parameter
XMuLV	xenotropic murine leukemia virus

Executive summary

This report presents the Stage 1 process characterization of the small virus retentive filtration step (Step 9) of the A-Mab drug substance process. The step removes virus by size. It is the dedicated small virus removal operation of the purification train and the principal source of clearance of minute virus of mice (MVM), and it also carries a claim for xenotropic murine leukemia virus (XMuLV). The study followed the characterization plan PCP-009 and supports the transfer of the process from Cambridge, MA, to Grafton, WI, under PTP-001. It was run on a scale-down model of the commercial step in two designed experiments in filtration volume (load) and filtration pressure, a screening design of 7 runs and a face-centred central composite design of 12 runs. Version 2.0 judges each output against the intermediate acceptance criteria of the master plan PCMP-001 Version 2.0. It analyses each response-surface output with the model that a model selection chose, in place of the full quadratic of the plan (Section 1.4).

Filtration volume was the only parameter with a detected effect. Across the range studied, from 50 to 140 L/m2, the mean MVM log reduction factor (LRF) fell by 1.73 log10 along a straight line. No term containing filtration pressure was significant for any output. The selected model of MVM LRF passes the model F-test of PCMP-001 §7.2, shows no lack of fit, and meets the limit on overfitting, which allows predicted R^2 to fall short of R^2 by at most 0.20. It fails the criterion on residuals and influential runs, because one run moves the fit more than that criterion allows. Its Cook's distance is above the median of the F distribution. That run, run 5, is the run at 50 L/m2 and 19 psi, with the highest LRF of the runs at that volume. XMuLV LRF has no adequate model and is judged on its measured results. Fragments and step yield did not depend on either parameter.

Each output with a criterion met its intermediate acceptance criterion across the normal operating range (NOR). The least favourable NOR combination is the combination of NOR low, set-point and NOR high of the two parameters that gives the lowest predicted LRF. There the mean predicted MVM LRF is 5.25 log10, against the LRF claimed for this step, 4.62 log10. Every measured XMuLV LRF, from 5.90 to 6.51 log10, met its claim of 4.62 log10, and XMuLV LRF has no PAR. For fragments the upper 95 % limit of a single run within the NOR is 0.78 %, against the criterion of 2.00 %. The proven acceptable range (PAR) of filtration volume for MVM is 50–136 L/m2 in the NOR-propagated analysis, in which the claim must hold with filtration pressure anywhere in its NOR. It extends 31 L/m2 beyond the upper NOR edge. The PAR of filtration pressure for MVM is the whole range studied. The design space covers 91 % of the characterized region on the grid evaluated, and only filtration volumes above 137.6 L/m2 lie outside it. With run 5 deleted, the PAR of filtration volume and the design space would both reach the end of the range studied, while the judgement over the NOR and the terms of the selected model would not change.

Both parameters were classified as well-controlled critical process parameters (WC-CPP). Filtration volume lowers the MVM LRF by 1.73 log10 across its range, and its PAR extends 31 L/m2 beyond its NOR. No effect of filtration pressure was detected, and its class follows the A-Mab case study and the scope of the viral clearance claim (Section 9). The integrated model of the train gives a process capability index (Cpk) of 1.51 for the cumulative MVM clearance of the drug substance and of 1.70 for XMuLV, both above the target of 1.33 and the minimum of 1.0 of PCMP-001. No simulated batch fell below either cumulative requirement.

Two study events were recorded, a transient pressure excursion to 22 psi and a membrane lot whose flux was 12 % below the value typical for the vendor. Every run concerned was retained in the analysis, as Version 1.0 recorded, and the record does not name the runs. The register of discrepancies holds 7 discrepancies from the plan (Section 13.2). Two concern the analysis, the model selection (PD-009-01) and the use of a model that does not meet the criterion on residuals and influential runs for the ranges and the design space (PD-009-02). Four record results and comparisons that PCP-009 required and the record does not hold (PD-009-03 to PD-009-06), among them the flux decay and the integrity tests of the qualification runs and the integrity tests and virus controls of the design runs. The last records the analyses that the plan does not contain, among them the deletion of every response-surface run in turn, a test of the curvature of the two designs against each other and the capability analysis (PD-009-07). The results of this report roll up into the master report PCMR-001, which consolidates the cumulative viral clearance claim.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose principal mechanism of action is antibody-dependent cell-mediated cytotoxicity (ADCC) (CMC Biotech Working Group 2009). It is purified by the platform train of Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small virus retentive filtration and ultrafiltration / diafiltration. The commercial process runs at a production bioreactor scale of 15,000 L.

The small virus retentive filtration step filters the flow-through pool of anion exchange through a membrane whose pore size distribution retains the parvoviruses while the antibody passes. The step is run at a set filtration pressure, in psi, up to a set filtration volume, which is the volume of load filtered per square metre of membrane, in L/m2. MVM, a parvovirus, is among the smallest viruses of concern, and this step is the principal source of its clearance. XMuLV, an enveloped retrovirus, is many times larger than the pores. The clearance of the step is orthogonal to that of low-pH viral inactivation (Step 6) and anion exchange chromatography (Step 8), because it removes virus by size and not by inactivation or by binding (International Council for Harmonisation 2023a).

The step forms no quality attribute. It contributes a log reduction to the two cumulative viral clearance requirements of the drug substance. Its filtrate is also judged against the drug substance criterion for fragments, because no later step changes them (Section 7). Step yield is a process performance output with no criterion.

The plan paired with this report is PCP-009. Table 1.1 lists the documents to which this report relates.

Table 1.1: Related documents.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-009	Process Characterization Plan (Small-Virus Retentive Filtration (Step 9))	Sibling — paired document

Document ID	Title	Relationship
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision
INV-002	Investigation Report (Response-surface models below the model-adequacy criteria of PCMP-001 Version 2.0)	Model adequacy of the response-surface designs

1.2 Objectives

The objectives of the study were those of PCP-009 §2.

1. Qualify a scale-down model of the step, and show that it represents the commercial step for virus retention, for flux decay and for step yield.
2. Determine which of the parameters affect the two viral clearance responses and the step yield, in what direction and by how much.
3. Fit a response-surface model for each response over the characterization ranges, and establish whether it predicts the mean response adequately.
4. Derive a proven acceptable range for each parameter against each of the two viral clearance responses.
5. Classify each parameter under SOP-4001 and define what the step contributes to the control strategy and to the modular viral clearance claim.

Version 2.0 of the plan adds fragments to objectives 2 to 4. The screening design identifies the parameters that have an effect. The response-surface model is the predictive model, and every range in this report comes from it. Section 12 states the outcome against each objective.

1.3 Regulatory and scientific basis

The study belongs to Stage 1, process design, of the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). It follows the enhanced approach of ICH Q8(R2) and ICH Q11. Prior knowledge and risk assessment identify the parameters that may affect quality, and designed experiments then measure and bound their effects (International Council for Harmonisation 2009, 2012). Criticality is treated as a continuum, as ICH Q8(R2) describes it, and the risk assessment that scoped the study followed ICH Q9(R1) (International Council for Harmonisation 2009, 2023b). Viral clearance is evaluated under ICH Q5A(R2). Each step that clears virus is studied in its own spiking model, and its log reduction enters the cumulative claim as a module (International Council for Harmonisation 2023a). The process model, the attribute framework and the risk method follow the A-Mab case study (CMC Biotech Working Group 2009).

1.4 Changes in this revision

Version 2.0 of this report replaces Version 1.0 of 2026-07-24. It is issued under the QA investigation INV-001, which was opened on 2026-08-03 after a review of the master report PCMR-001 for readiness of the Biologics License Application, and closed on 2026-09-18 with the classification major. INV-001 found that the criteria used to judge each step in the PAR analysis had been set after the data were seen and were in no approved protocol (INV-001 finding 1). The Version 1.0 limits of this step were among them. They were applied at analysis and appear in no approved plan. This revision also applies the second investigation INV-002 (opened 2026-08-31, closed 2026-09-28, classification minor), which applied the model-adequacy criteria of PCMP-001 §7.2 to every response-surface model with a criterion. The changes to this report are the following.

1. Acceptance criteria. MVM LRF and XMuLV LRF are each judged against the platform claim of this step, 4.62 log10, under PCMP-001 §7.4 (Table 1.2). Version 1.0 judged them against the rule “nominal claim less 0.35 log10 for assay variability”, which gave 4.97 for MVM and 5.47 for XMuLV. Fragments are added as an output, with the drug substance criterion of 2.00 %.
2. Fragment method. Fragments are measured by non-reduced capillary electrophoresis with sodium dodecyl sulfate (nrCE-SDS), the method that the analytical method decision AMD-001 of 2026-03-20 set for every study at every step. Both designs of this step started after that date, so every run of both has an nrCE-SDS result (Table 3.3). Version 1.0 did not assess fragments at this step.
3. Statistic that decides the verdict. Viral clearance is judged on the mean predicted LRF at the least favourable combination of NOR low, set-point and NOR high of the parameters (PCMP-001 §6.4). Fragments are judged on the 95 % prediction interval of a single run. Version 1.0 judged every output on a 95 % interval for a single batch from a Monte-Carlo simulation of the fitted model.
4. Models. Each response-surface output is analysed by the model selection that INV-002 adopted as its corrective action CA3 on 2026-09-15, in place of the full quadratic model that the plan prescribed (Section 3.5). Under it XMuLV LRF has no adequate model and is judged on its measured results, and the selected model of MVM LRF does not meet the criterion on residuals and influential runs (Section 5.3).
5. Scale-down model. The model is qualified by its offset from manufacturing scale (PCMP-001 §7.1) and not by a test of equivalence (Section 3.1).
6. Record. The execution window of every study is now given (Table 4.2), which the Version 1.0 records did not give (INV-001 finding 5). The departures from the plan that the re-analysis found are recorded as discrepancies from the plan (Section 13.2).

Table 1.2: Acceptance criteria for the step outputs in Version 1.0 and Version 2.0, virus filtration.

Attribute	Version 1.0[a)]		Version 2.0	
	Criterion	Rule	Criterion	Basis
Fragments, LMW (%)	not assessed[b)]	–	≤ 2.00	Drug substance
MVM clearance (log10)	≥ 4.97	R4	≥ 4.62	Platform claim
XMuLV clearance (log10)	≥ 5.47	R4	≥ 4.62	Platform claim

[a)] Applied in the Version 1.0 reports at analysis. No approved Version 1.0 plan contained these criteria (INV-001 finding 1).
[b)] Version 1.0 did not report fragments. The attribute was added at Version 2.0.
[R4] Version 1.0 rule: nominal claim less 0.35 log10 for assay variability.

The data, the designs, the NOR and the set-points of the step did not change. Recomputed under the Version 1.0 criteria, statistic and model, with each edge rounded inward as in Table 7.3, the NOR-propagated PAR of filtration volume for MVM is 50–97.2 L/m2 and that of filtration pressure 12.2–24.7 psi, and neither contains the NOR. Under Version 2.0 they are 50–136 L/m2 and 8–30 psi, and both contain it (Section 7). Version 1.0 also gave XMuLV LRF PARs of 50–140 L/m2 and 8–30 psi from its full quadratic model. These PARs are withdrawn, because XMuLV LRF has no adequate model and therefore no PAR.

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The small virus retentive filtration step has been used to manufacture three earlier licensed humanized IgG1 antibodies of the platform, X-Mab, Y-Mab and Z-Mab, and to supply A-Mab clinical material, with the same filter type and the same mode of operation (CMC Biotech Working Group 2009). Their history gives the platform claim of the step, 4.62 log10 for each of the two model viruses (PCMP-001 §7.4). The mechanism of removal is size exclusion and does not depend on the antibody, so the platform experience gives the expected direction of each effect. How the anion exchange pool of A-Mab fouls the membrane is specific to the product, and the effects were therefore measured on A-Mab material.

MVM sits close to the pore size cut-off of the membrane. Its retention depends on the fraction of pores at the large end of the distribution and on the flow through them. As the membrane fouls with product and residual impurity, the flux shifts toward the least fouled and largest pores, and small virus can pass. MVM retention is therefore expected to fall as the volume filtered per unit area rises, so the claim is most at risk at the maximum load. PCP-009 §6.3 also expected the fall to accelerate toward the top of the volume range, as the shift of flux to the largest pores progresses.

A higher filtration pressure raises the flux. It also compacts the fouling layer, which raises the resistance of that layer, but the flux still rises with pressure. PCP-009 expected MVM retention to fall at high pressure as more filtrate passes the largest open pores. A low pressure or an interruption of flow lets virus held in the pore structure diffuse through. For this reason the claim of the step is made over a pressure range and for uninterrupted filtration. PCP-009 §4.3 reasoned that the load decides how much of the membrane is still open and the pressure how fast the filtrate crosses what is left. On that reasoning the plan expected the fall of MVM retention with load to be steeper at high pressure than at low pressure.

XMuLV is many times larger than the pores. It is retained while the membrane is intact, so its clearance is expected not to depend on either parameter. The antibody passes the pores freely, so step yield loses only the product held up in the system and in the fouling layer on the membrane. The hold-up of the system is fixed, so it is a smaller share of a larger load and on its own would raise the yield as the load rises. The fouling layer, and the product held in it, grow with the volume filtered and lower the yield. PCP-009 expected yield to be high and to fall slowly as the load rises. That holds only if the loss in the fouling layer grows faster than the share of the fixed hold-up falls. Fragments, the clipped species of the antibody, are smaller than the intact antibody and pass the pores in the same way, so the step is not expected to change the fragment level of its load.

2.2 Quality attributes in scope

The step contributes log reduction to the two viral safety attributes of Table 2.1 and is judged against the criterion for fragments. The acceptance values of the two viral safety attributes are cumulative requirements on the drug substance. They are met by the claims of several steps together and cannot be applied to one step. PCMP-001 §3 takes both requirements from the A-Mab case study. There they are the total log reductions of the platform train, low-pH treatment, anion exchange and small virus retentive filtration together, for A-Mab and three other antibodies (CMC Biotech Working Group 2009, 165). For XMuLV the case study checks that total against the retrovirus-like particles per dose estimated by electron microscopy (CMC Biotech Working Group 2009, 166). It gives this check for XMuLV only.

Table 2.1: Quality attributes to which the step contributes, with their drug substance criteria and criticality.

Category	CQA	Acceptance	Criticality	Tool #1
viral safety	Viral clearance — MVM (parvovirus)	≥ 8.6 log10 (cumulative)	VH	20
	Viral clearance — XMuLV (enveloped)	≥ 16.7 log10 (cumulative)	VH	20
size variant	Fragments (LMW, clipped species)	≤ 2 % (nrCE-SDS)	L	12

VH, very high. L, low. Tool #1 is the criticality score of the A-Mab case study, the product of impact and uncertainty.

Both viral safety attributes are of very high criticality. It was assigned by rule, because a viral contamination would harm the patient directly (PCMP-001 §3). MVM is the virus whose retention the operating parameters of this step can degrade. Low-pH viral inactivation makes no claim for MVM, because a parvovirus has no lipid envelope for low pH to disrupt. The cumulative MVM requirement of 8.6 log10 therefore rests on the claims of anion exchange and of this step alone (PCMP-001 §7.4).

Fragments are of low criticality. They form in culture and during the harvest hold, where proteases released from lysed cells clip the antibody, and the CQA register assigns them to harvest. The A-Mab case study lists fragmentation among the typical quality attributes of an antibody, but its table of the basis for acceptable ranges gives fragments no limit (CMC Biotech Working Group 2009, 30, 56). The sponsor set the criterion of 2.00 % when fragments joined the CQA register at Version 2.0 (PCMP-001 §3).

2.3 Risk-based prioritization and parameter selection

The risk assessment RA-001 ranked every parameter of the train twice, for its potential impact on the quality attributes and for its potential to interact with other pa-

rameters (International Council for Harmonisation 2023b). Parameters ranked high on either ranking were assigned to multivariate study, and parameters ranked lower to univariate study. For this step RA-001 assigned its 2 parameters to multivariate study and none to univariate study. Filtration volume was ranked high because fouling shifts the flow to the larger pores as the volume filtered per unit area rises, so MVM retention falls. Filtration pressure was ranked high because a loss of pressure or a stop of flow lets MVM held in the pore structure diffuse into the filtrate, and because a higher pressure drives more filtrate through the pores that remain open. No parameter of the step was left out of the study on prior knowledge alone, because a viral safety claim is supported by product specific data even where the mechanism is well understood.

3 Materials and methods

3.1 Scale-down model and its qualification

The scale-down model uses the same filter type as manufacturing, and the load is scaled by membrane area (PCMP-001 §5). The volumetric load in L/m2 and the filtration pressure of the model therefore correspond to the commercial step.

The qualification runs were made in study S-VF-SDM, from 2026-04-15 to 2026-04-24, under the scale-down model procedure SOP-1001 and before the screening runs began. The study compared 3 scale-down runs at the set-points with 5 manufacturing-scale batches. PCP-009 §5.1 named three outputs for this comparison, the flux decay profile, the step yield and the post-use integrity test result, and required the load material, the filtration temperature and the hold-up volume per area to be matched. The record holds step yield alone. The flux decay profile and the post-use integrity test results are not recorded for either scale, and neither is the matching (PD-009-03). Version 1.0 of this report stated that every run passed its post-use integrity test, but no result of the test is recorded, for these runs or for the design runs (Section 3.2). The comparison therefore rests on step yield alone.

The model is qualified by the offset rule of PCMP-001 §7.1. The offset of an output is the mean of the scale-down runs at the set-points minus the mean of the manufacturing batches, with its two-sided 90 % confidence interval. The model is qualified for an output when the conclusion against its criterion over the NOR is the same with the offset at either end of that interval. Step yield has no criterion at this step, so its offset is recorded and no conclusion depends on it (Table 3.1). The mean step yield was 97.6 % at scale-down and 98.1 % at manufacturing scale, an offset of −0.5 percentage points with a 90 % confidence interval of −1.2 to 0.3 points.

Table 3.1: Offsets of the scale-down models (scale-down minus manufacturing scale) and the qualification of each model for each output, virus filtration.

Output	Runs		Mean		Offset		Met over NOR (low/est./high)	Qualified
	SDM	Mfg	SDM	Mfg	Estimate	90 % CI		
Step yield (%)	3	5	97.6	98.1	−0.5	−1.2 to 0.3	no criterion[a)]	n.a.[a)]

[a)] The output has no acceptance criterion at this step, so no conclusion depends on its offset and qualification does not apply. The offset is recorded.
SDM, scale-down model. Mfg, manufacturing scale. Offset, scale-down mean minus manufacturing mean. Met over NOR: the criterion is met at every combination of NOR low, set-point and NOR high (the NOR in force), with the offset at the low end of its interval, at its estimate (est.) and at the high end.

The two LRF and fragments were not compared at the two scales. Their offset is unknown, not zero, and they are judged as the scale-down model gives them (PCMP-001 §7.1). Virus cannot be spiked into a manufacturing batch, so every clearance figure of this report comes from the scale-down model, and its representativeness rests on the spiking model (International Council for Harmonisation 2023a). The representativeness of the fragment results rests on the mechanism of the step (Section 2.1). No output with a criterion therefore has a measured offset, and the judgements of MVM LRF, XMuLV LRF and fragments are made on the scale-down results under these assumptions.

3.2 Operation

PCP-009 §5.2 specifies the flow-through pool of anion exchange as the load of every run, and the record does not identify the pool used in each run (PD-009-06). Each design point was run once, at the filtration pressure of the point and up to its filtration volume (Appendices A and B). Both model viruses were spiked into the load of every run, and every run gave an MVM LRF and an XMuLV LRF. The order of execution of each run is recorded and enters the residual tests (Section 3.5). The designs varied the two parameters of Table 4.1. The other operating variables of the step are set by the operation procedure SOP-2012 and were not factors of the designs.

PCP-009 §5.2 planned the step yield of each design run from the same spiked run, by protein mass balance from the ultraviolet absorbance of the load and of the pooled filtrate. PCP-009 §5.4 asked instead for an unspiked load and an unspiked filtrate pool to be assayed for the yield, which cannot exist when every design run is spiked, so the plan conflicts with itself. The record gives the yield of each spiked run as a single value. It holds no protein concentration, no volume, no named assay and no sample of the load taken before spiking (PD-009-05).

PCP-009 §5.2 required the filter of each run to be integrity tested before and after the run under SOP-2012, and a run whose post-use test failed to be excluded. No integrity test result is recorded for any design run (PD-009-04), and no run was excluded. The claim made from these runs applies to uninterrupted filtration, and no run of the record is described as interrupted.

3.3 Analytical methods

The methods of Table 3.2 measured the outputs. MVM was titred by TCID50 with qPCR confirmation (AMV-3018), XMuLV by TCID50 (AMV-3017) and fragments by nrCE-SDS (AMV-3020). The LRF of a run is the base-ten logarithm of the total virus in the spiked load divided by the total virus in the filtrate (PCP-009 §5.3). PCP-009 §5.2 names ultraviolet absorbance for the step yield, and the record names no assay (Section 3.2). Appendix D summarises the methods.

Table 3.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-1002	Design, Execution and Statistical Analysis of DoE Studies
	SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
Analytical method validation	AMV-3018	MVM Infectivity Titre (TCID50/qPCR)
	AMV-3017	XMuLV Infectivity Titre (TCID50)
	AMV-3020	Purity (non-reduced CE-SDS)

The record holds one LRF per run for each virus. It does not hold the titres of the spiked load and the filtrate, the limit of detection reached, or the results of the cytotoxicity, interference and spike recovery controls that PCP-009 §5.3 required. Nor does it hold the achieved titre of each run, the clarification of the virus stock or the identification of each sample by run and volumetric load (PCP-009 §5.4 and §12). Every LRF in the record is a point value. Whether any filtrate titre was below the limit of detection, which would make its LRF a lower bound, cannot be read from the record.

Both designs ran after the method decision AMD-001, so fragments were measured by nrCE-SDS on every run of both designs (Table 3.3). The method has a limit of quantitation of 0.1 % and an absolute standard deviation of 0.06 percentage points. Aggregate was not measured at this step.

Table 3.3: Methods used for size variants in the designed studies, virus filtration.

Attribute	Method used in study	
	Screening	Response surface
Aggregates, HMW	not measured[a)]	
Fragments, LMW	nrCE-SDS	nrCE-SDS

[a)] Aggregate was not measured in the designs at this step.

3.4 Sampling plan

PCP-009 §5.4 planned samples of the load before spiking, of the spiked load and of the filtrate, with the filtrate sampled as sequential fractions across the volumetric load and as a pool. The record holds one result per run for each output, no result for a fraction and not the volume actually filtered in each run (Section 13.2). The log reduction therefore cannot be resolved against the volume filtered within a run, and the effect of filtration volume is estimated between runs only. The screening design

had 3 replicates of the centre point, and the response-surface design had 4. Their spread gives the pure error of each design for the lack of fit tests.

3.5 Statistical methods

Both factors were coded to −1 and +1 at the ends of the range studied and to 0 at its midpoint, so that coefficients are comparable between the two factors and between the two designs. A coded coefficient is the change in the response over half the range studied, 45 L/m2 for filtration volume and 11 psi for filtration pressure. Terms significant at $\alpha = 0.05$ were treated as real (PCMP-001 §6.2). In this report the terms of a model are counted without the intercept.

The screening design was analysed by ordinary least squares with both main effects and their interaction. An effect is twice the coded coefficient, which is the change in the response across the whole range studied, as PCMP-001 §6.2 states it. PCP-009 §5.5 states effects as the change over half the range, which is the coefficient, and Appendix C gives both. The screening model identifies effects. Its centre points test for curvature through the one degree of freedom left for lack of fit, but they cannot attribute the curvature to a factor.

Each response-surface output was analysed by model selection, the analysis that INV-002 adopted as corrective action CA3. It replaced the full quadratic model of PCP-009 §5.5 (Section 13.2). The selection has three steps.

1. The scale. The Box-Cox profile of the full quadratic gives the power of the response that best suits a least squares fit, with its 95 % interval (Box and Cox 1964). The log scale is used where that interval contains 0 and excludes 1, and the raw scale otherwise. Every interval at this step contained both 0 and 1, so every output was modelled on the raw scale.
2. The terms. The full quadratic in two factors has 5 terms, the two main effects, their interaction and the two squares. It has 13 hierarchical submodels, counting the intercept-only model, in which an interaction appears only with both of its main effects and a square only with its main effect. Every submodel was fitted, and the one with the lowest AICc was selected. AICc estimates how well a model would predict new runs and penalises every estimated term. Its small-sample correction raises the penalty when the runs are few for the number of terms, as in these designs (Hurvich and Tsai 1989; Burnham and Anderson 2002). The residual variance is counted as a parameter of the model. A stepwise search changes one term at a time and can stop where only a combination of changes would lower AICc further. On these data a stepwise search in either direction would have stopped at the same model for every output with a criterion.
3. The check. The selected model was judged against the four criteria of PCMP-001 §7.2 below. An intercept-only model states that the output does not depend on the parameters within the range studied. An output with no adequate model is judged on its measured results and sets no range.

PCP-009 §7 states the four adequacy criteria of PCMP-001 §7.2 with their thresholds.

1. A significant model. The model F-test is significant at $\alpha = 0.05$. An intercept-only model has no F-test, and the criterion does not apply to it.
2. No significant lack of fit. The lack of fit test against the pure error of the centre points is not significant at $\alpha = 0.05$.
3. The limit on overfitting. Predicted R^2, calculated from the prediction error sum of squares, falls short of R^2 by no more than 0.20 (Montgomery 2017).
4. Residuals and influence. The absolute residuals regressed on the predicted values, and the residuals regressed on the order of execution, show no slope significant at $\alpha = 0.05$. A run with a Cook's distance above 4/n is examined by deleting it and repeating the judgement. The model fails the criterion only where a run's Cook's distance exceeds the median of the F distribution with p and $n - p$ degrees of freedom, where p is the number of model parameters including the intercept and n the number of runs.

R^2, adjusted R^2 and predicted R^2 are reported for every model. PCP-009 §7 also keeps the Version 1.0 conditions that the residuals show no material departure from normality and that the residual standard error is small against the distance between the predicted response and its criterion, and it leaves the judgement of both to this report. Normality is judged on the normal quantile plot and by the Shapiro-Wilk test at $\alpha = 0.05$. The residual standard error is judged small against that distance when the end of the 95 % prediction interval of a single run on the side of the criterion still meets the criterion at the least favourable NOR combination. Several analyses of this report are not in PCP-009 (PD-009-07). They are the deletion of every response-surface run, the test of the curvature of the two designs against each other, the Shapiro-Wilk test and the reading of the two Version 1.0 conditions by the prediction interval of a single run, the comparisons with the models within 2 AICc units of the selected one and with stepwise searches, and the capability analysis.

Each output with a criterion was judged at every combination of NOR low, set-point and NOR high of the two parameters, 9 combinations in all. Viral clearance was judged on the mean predicted LRF at the least favourable of these combinations, because viral clearance is validated at worst-case conditions (PCMP-001 §6.4). The lower end of the two-sided 95 % prediction interval of a single run there is given for information. Fragments were judged on the exact 95 % prediction interval of a single run at each combination, which includes the uncertainty of the fitted coefficients and the residual. The interval is two-sided, so its upper end is a one-sided 97.5 % limit. An output with no adequate model was judged on every measured result of both designs. No offset was applied, because no output with a criterion has a measured offset at this step (Section 3.1). Section 7 defines the two PAR analyses, and Section 6 the grid on which the design space was evaluated.

Capability was estimated from the integrated model of the train (PCMP-001 §6.5) and judged against the target and the minimum of PCMP-001 §7.9. The analyses followed the design of experiments procedure SOP-1002. The Monte-Carlo analyses use a fixed random seed, so every number of this report can be regenerated from the data.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 gives the two parameters of the step with their set-points, their NOR and the ranges studied. Both parameters were factors of both designs. The NOR and the set-points were not changed at Version 2.0, so the NOR in force is the NOR as characterized.

Table 4.1: Parameters of the step, with their set-points, normal operating ranges and the ranges studied.

Code	Parameter	Unit	Set-point	NOR	Range studied
A	Filtration volume (load)	L/m2	90	50–105	50–140
B	Filtration pressure	psi	13	10–20	8–30

Code, the letter of the factor in the design matrices and the effect tables.

The range of filtration volume runs from the lower NOR edge, 50 L/m2, to 140 L/m2, above the upper NOR edge of 105 L/m2. It was widened upward only, because MVM retention was expected to fall as the load rises, so that the edge of acceptable performance would lie above the NOR. The pressure range of 8–30 psi extends beyond the NOR of 10–20 psi on both sides. Low pressures were included because a low pressure was expected to let retained virus diffuse through, and high pressures because they drive more filtrate through the largest open pores. The ranges studied form the knowledge space of the step, and no claim in this report extends beyond them.

The studies of the step ran one after the other (Table 4.2). Each window runs from the first to the last day of runs of its study, and all of them closed before Version 1.0 of this report was approved on 2026-07-24. The record names no instrumented system under calibration and change control for these studies (PCMP-001 Table 5.1), so no calibration status is given.

Table 4.2: Execution windows of the studies of the step.

Study	Study type	Start	End
S-VF-SDM	Qualification of the scale-down model	2026-04-15	2026-04-24
S-VF-DOE	Screening design	2026-04-27	2026-05-05
S-VF-RSM	Response-surface design	2026-05-06	2026-05-15

4.2 Screening design

The screening study S-VF-DOE was a two-level full factorial in both factors with centre points, 7 runs in all, of which 4 were the factorial runs at the corners of the ranges studied and 3 were replicates of the centre point (Appendix A). With two factors the full factorial estimates both main effects and their interaction free of aliasing. The interaction model fits four parameters to 5 distinct design points, the corners and the centre, so its fit statistics overstate how well it fits (PCMP-001 §6.2). The one degree of freedom left for lack of fit tests for curvature.

4.3 Response-surface design

The response-surface study S-VF-RSM was a face-centred central composite design in both factors, 12 runs in all, made up of 4 factorial runs, 4 axial runs and 4 replicates of the centre point (Appendix B). The axial runs sit on the edges of the ranges studied, because settings outside those ranges are not settings for which the step may be claimed. The design supports the full quadratic model in both factors. Its centre points give the pure error for the lack of fit test, with 3 degrees of freedom.

4.4 Univariate assessment

No parameter of this step was assigned to univariate study. PCP-009 §6.4 allowed a univariate confirmation run where the designs could not separate an effect, and none was made.

5 Results

5.1 Centre-point performance and reproducibility

Table 5.1 gives the centre-point results of both designs. In the response-surface design the SD of the four centre points was 0.12 log10 for MVM LRF and 0.18 log10 for XMuLV LRF. The three MVM centre points of the screening design spread more widely, from 5.01 to 5.97 log10, an SD of 0.49 log10. The variance of the centre points of each design is its pure error, the basis of its lack of fit test.

Table 5.1: Centre-point results of the two designs.

Design	Response	n	Mean	SD	CV (%)
Screening	MVM LRF (log10)	3	5.44	0.49	9.0
	XMuLV LRF (log10)	3	6.30	0.15	2.3
	Fragments, LMW (%)	3	0.61	0.02	3.9
	Step yield (%)	3	98.6	0.77	0.8
Response surface	MVM LRF (log10)	4	5.54	0.12	2.1
	XMuLV LRF (log10)	4	6.13	0.18	2.9
	Fragments, LMW (%)	4	0.67	0.09	12.9
	Step yield (%)	4	98.3	0.84	0.9

SD and mean in the unit of the response. CV, the SD as a percentage of the mean.

5.2 Factor effects in the screening design

Filtration volume had the only significant effect in the screening design (Table 5.3). Raising it from 50 to 140 L/m2 lowered MVM LRF by 1.61 log10 ($p = 0.027$). The effect of pressure on MVM LRF (-0.25 log10, $p = 0.581$) and the interaction (0.02 log10, $p = 0.965$) were not significant. The F-test of the whole screening model of MVM LRF gave $p = 0.096$, with 3 residual degrees of freedom. No term was significant for XMuLV LRF, fragments or step yield.

Table 5.2: Screening models of the four responses.

Response	R^2	Adj. R^2	Pred. R^2	Model F-test p	Curvature p
MVM LRF	0.85	0.70	0.77	0.096	0.935
XMuLV LRF	0.71	0.43	0.36	0.236	0.847
Fragments, LMW	0.09	−0.83	−30.57	0.958	0.0497
Step yield	0.71	0.42	−2.23	0.242	0.486

Curvature p, the lack of fit test of the interaction model, one degree of freedom, which compares the centre points with the factorial runs.

Table 5.3: Effects in the screening design, as the change in the response across the range studied.

Response	Term	Effect	Std. err.	p-value
MVM LRF (log10)	A	−1.61	0.40	0.027
	B	−0.25	0.40	0.581
	AB	0.02	0.40	0.965
XMuLV LRF (log10)	A	−0.08	0.12	0.548
	B	0.26	0.12	0.120
	AB	0.19	0.12	0.220
Fragments, LMW (% points)	A	0.01	0.06	0.832
	B	0.03	0.06	0.687
	AB	0.01	0.06	0.869
Step yield (% points)	A	0.57	0.73	0.494
	B	0.54	0.73	0.514
	AB	1.81	0.73	0.089

A, filtration volume (load). B, filtration pressure. AB, their interaction. Std. err., the standard error of the effect.

The centre points of the screening design showed curvature for fragments. The three centre points read 0.08 percentage points lower than the factorial runs, which is significant ($p = 0.0497$). In the response-surface design the four centre points read 0.05 points higher than the factorial runs, the opposite sign. The difference between the two estimates, 0.13 points, is not significant ($p = 0.132$), by a t-test on the pooled SD of the centre points of both designs with 5 degrees of freedom. In size both estimates are at most 0.08 percentage points, against a criterion of 2.00 %, and of the order of the SD of the method, 0.06 points. No other response showed curvature in the screening design ($p \geq 0.48$). The screening results identify the effects, and the response-surface design gives the predictive model.

5.3 Response-surface models

Table 5.4 compares the full quadratic model that PCP-009 planned, fitted to the 12 planned runs, with the model each output now uses. The full quadratic did not meet

the limit on overfitting for any output with a criterion. Its R^2 exceeded its predicted R^2 by 0.42 for MVM LRF and by more for XMuLV LRF and fragments. Its adjusted R^2 was 0.90 for MVM LRF, 0.09 for XMuLV LRF and −0.35 for fragments. INV-002 named a refit as the remedy for each of them. Viral clearance is not a drug substance test, so no further run was spiked with virus, and fragments had no effect to resolve. The step was therefore not augmented.

Table 5.4: Re-evaluation of the response-surface models.

Attribute	Planned runs, full quadratic				Model used (terms)
	Runs	R^2	Pred. R^2	Gap	
MVM clearance (log10)	12	0.95	0.53	0.42	Filtration volume (load)
XMuLV clearance (log10)	12	0.51	−1.14	1.65	None[a)]
Fragments, LMW (%)	12	0.27	−1.74	2.01	No effect

[a)] No adequate model. The output is judged on its measured results. Gap: R^2 minus predicted R^2, at most 0.20 under PCMP-001 §7.2. Model used: the scale chosen by the Box-Cox profile and the model with the lowest AICc among the hierarchical submodels of the full quadratic, on one block of runs (INV-002).

The model selection gave a straight line in filtration volume for MVM LRF, a model in pressure alone for XMuLV LRF, and an intercept-only model for fragments and for step yield. Table 5.5 judges each selected model against the model F-test, the lack of fit test and the limit on overfitting of PCMP-001 §7.2, and Table 5.6 against the criterion on residuals and influential runs.

Table 5.5: Selected models against the model F-test, the lack of fit test and the limit on overfitting of PCMP-001 §7.2.

Output	Terms	R^2	Adj. R^2	Pred. R^2	Residual SD	Model F-test		Lack of fit		Limit on overfitting	
						p	Met	p	Met	Gap	Met
MVM LRF	A	0.90	0.89	0.84	0.229	< 0.001	Met	0.105	Met	0.06	Met
XMuLV LRF	B	0.32	0.26	0.08	0.143	0.053	Not met	0.789	Met	0.24	Not met
Fragments, LMW	none	0.00	0.00	−0.19	0.062	n.a.	n.a.	0.911	Met	0.19	Met
Step yield	none	0.00	0.00	−0.19	0.588	n.a.	n.a.	0.923	Met	0.19	Met

A, filtration volume. B, filtration pressure. Residual SD in the unit of the output, log10 for the LRF and percentage points for fragments and step yield. Gap, R^2 minus predicted R^2, at most 0.20. n.a., not applicable, an intercept-only model has no F-test.

Table 5.6: Selected models against the criterion on residuals and influential runs of PCMP-001 §7.2.

Output	Residual trend p		Cook's distance				Criterion met
	Absolute residuals	Order	Largest	F median	Runs above 4/n	Runs above F median	
MVM LRF	0.061	0.365	0.79 (run 5)	0.74	2, 5	5	Not met
XMuLV LRF	0.166	0.362	0.33 (run 3)	0.74	none	none	Met
Fragments, LMW	n.a.	0.961	0.47 (run 9)	0.49	9	none	Met
Step yield	n.a.	0.359	0.32 (run 11)	0.49	none	none	Met

Absolute residuals, the regression of the absolute residuals on the predicted values, not tested for an intercept-only model. Order, the regression of the residuals on the order of execution. 4/n is 0.33 with n = 12 runs. F median, the median of F(p, n − p), which depends on the number of parameters p.

MVM LRF is modelled by a straight line in filtration volume (Table 5.7). The Box-Cox interval of λ contains both 0 and 1, so the raw scale was kept. On the grid of the selection the interval runs from the lower end of the grid, −2.00, to 1.99. On a grid of λ from −10 to 10 it runs from −5.46 to 1.99. The intercept, 5.44 log10, is the mean predicted LRF at the midpoint of the volume range, 95 L/m2. The coefficient of filtration volume is −0.866 log10 per coded unit, with a standard error of 0.093. Across the range studied the mean LRF therefore falls by 1.73 log10, which is 0.19 log10 for every 10 L/m2. The model F-test is significant ($p < 0.001$), the lack of fit is not ($p = 0.105$, Table 5.8), and R^2 exceeds predicted R^2 by 0.06. The model therefore passes the F-test, shows no lack of fit and meets the limit on overfitting. The residual SD of the model is 0.23 log10, against an SD of the centre points of 0.12 log10. The residuals show no departure from normality (Shapiro-Wilk $p = 0.910$, Figure 5.1). The next model by AICc adds the main effect of pressure and lies 4.6 units higher, so no other model is within 2 units of the selected one.

Table 5.7: Selected model of MVM LRF, coefficients in coded units.

Term	Coefficient	Std. err.	t	p-value
Intercept	5.439	0.066	82.4	< 0.001
A	−0.866	0.093	−9.3	< 0.001

A, filtration volume, coded −1 to +1 over the range studied. Coefficients in log10.

Table 5.8: Analysis of variance of the selected model of MVM LRF, with the lack of fit against pure error.

Source	Sum of squares	df	Mean square	F	p-value
Model	4.499	1	4.499	86.1	< 0.001
Residual	0.522	10	0.052		
Lack of fit	0.482	7	0.069	5.1	0.105
Pure error	0.041	3	0.014		

Lack of fit and pure error partition the residual.

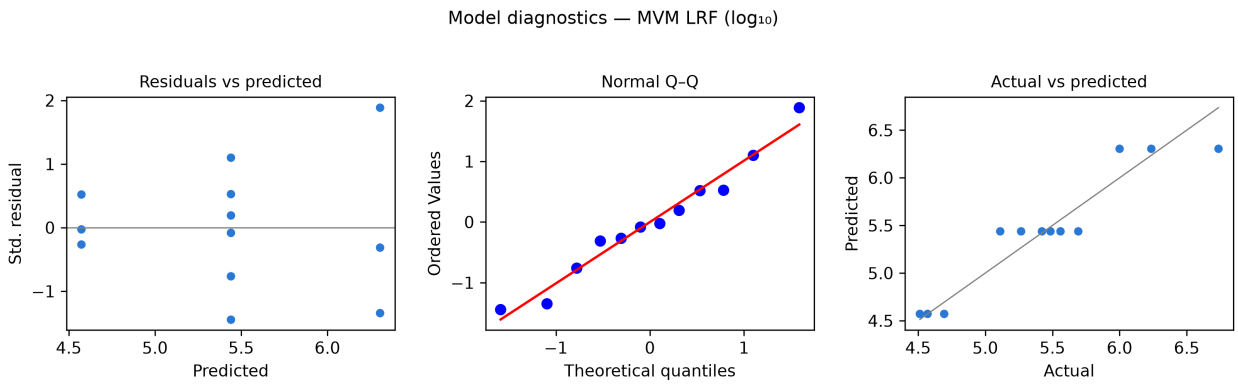


Figure 5.1: Diagnostics of the selected model of MVM LRF, residuals against predicted values, normal quantile plot, and actual against predicted values.

The model of MVM LRF does not meet the criterion on residuals and influential runs. Neither residual trend is significant (p = 0.061 for the absolute residuals against the predicted values and p = 0.365 for the residuals against the order of execution). Runs 2 and 5 have a Cook's distance above 4/n = 0.33. Run 5, at 0.79, also exceeds the median of F(p, n − p), 0.74, and so fails the criterion. Run 5 is an axial run at the lowest filtration volume, 50 L/m2, and the centre pressure, 19 psi, executed 10th of 12. Its LRF, 6.74 log10, is the highest of the 3 runs of the design at that volume. Its residual is 0.43 log10, and its externally studentized residual 2.9. Its leverage, 0.25, is below 2p/n = 0.33. Run 2, a factorial run at the lowest volume and the highest pressure, has a residual of −0.31 log10.

Both flagged runs were deleted in turn, the model selection was repeated and MVM LRF was judged over the NOR again (Table 5.9). Every other run of the design was deleted in the same way, because the record does not name the runs of the two study events (Section 13.1). Every deletion leaves filtration volume as the only term of the selected model and the mean predicted LRF at the least favourable NOR combination between 5.20 and 5.27 log10, above the claim of 4.62 log10. The deletion of run 5 flattens the line, from −0.866 to −0.770 log10 per coded unit, and the mean prediction then meets the claim over the whole range studied. The NOR-propagated PAR of filtration volume then reaches the end of the range studied, and so does the design space. The deletion of run 2 steepens the line to −0.934 and moves the point at which the mean prediction crosses the claim from 137.6 to 136.1 L/m2. With run 3 or run

7 deleted, the Box-Cox interval of the remaining runs excludes 1, so that the model is fitted on the log scale, and the deletion of run 7 also carries the PAR edge to the end of the range. Across the single deletions the upper PAR edge of filtration volume lies between 135 L/m2 and the end of the range studied. No deletion changes the judgement over the NOR, and the NOR lies 30 L/m2 or more below the edge under every deletion. The model of all runs, which gives a narrower range of volume than the model without run 5, was kept for the ranges and the design space (Section 13.2).

Table 5.9: Selected model of MVM LRF with each response-surface run deleted in turn, and the judgement over the NOR.

Run deleted	Run type	Cook's distance	Scale	Mean LRF at the least favourable NOR combination	Upper edge of the PAR of volume (L/m2)	Design space (% of grid)
none	–	–	raw	5.25	136	91
1	factorial	0.02	raw	5.25	136	91
2[a)]	factorial	0.40	raw	5.27	135	91
3	factorial	0.06	log	5.20[b)]	136	91
4	factorial	0.00	raw	5.25	136	91
5[c)]	axial	0.79	raw	5.22	140 (end)	100
6	axial	0.02	raw	5.26	137	91
7	axial	0.10	log	5.24[b)]	140 (end)	100
8	axial	0.03	raw	5.26	137	91
9	centre	0.00	raw	5.24	136	91
10	centre	0.01	raw	5.24	136	91
11	centre	0.06	raw	5.22	135	91
12	centre	0.00	raw	5.25	136	91

[a)] Cook's distance above 4/n = 0.33, examined by deletion.
[b)] The median, because the model of the remaining runs is on the log scale.
[c)] Cook's distance above the median of F(p, n − p) = 0.74, the limit of the criterion.
Every deletion keeps filtration volume as the only term, and every predicted LRF in the table meets the claim of 4.62 log10. Scale log, the Box-Cox interval of the remaining runs excludes 1. PAR, the NOR-propagated range, its edge rounded inward to three significant figures. End, the PAR reaches the end of the range studied.

XMuLV LRF has no adequate model. The selected model carries pressure alone, with a coefficient of 0.128 log10 per coded unit and a residual SD of 0.14 log10, but its F-test is not significant (p = 0.053) and R^2 exceeds predicted R^2 by 0.24, above the limit. XMuLV LRF is therefore judged on its measured results, all 19 of which, from 5.90 to 6.51 log10, met the claim of 4.62 log10. The intercept-only model lies 1.0 AICc units from the selected one, shows no lack of fit and meets the limit on overfitting. It predicts a mean LRF of 6.20 log10 everywhere, and the judgement over the NOR would be the same under it.

The selected model of fragments is intercept-only, with a mean of 0.64 % and a residual SD of 0.06 percentage points. The lack of fit is not significant (p = 0.911), and the model meets the limit on overfitting with a gap of 0.19. Its residuals show no departure from normality (Shapiro-Wilk p = 0.744), and the upper 95 % limit of a single run, 0.78 %, meets the criterion of 2.00 %. Both conditions kept from Version 1.0, normal residuals and a residual SD small against the distance to the criterion, are therefore

met. For an intercept-only model R^2 is zero and predicted R^2 is $1 - (n/(n-1))^2$, so the gap, $(2n-1)/(n-1)^2$, is fixed by the number of runs. It is 0.19 with 12 runs and 0.21 with 11. Run 9, a centre run, has a Cook's distance of 0.47, above 4/n and below the median of F(p, n − p), 0.49. The deletion of run 9, or of any other single run, leaves an intercept-only model on 11 runs, whose gap of 0.21 exceeds the limit on overfitting whatever the results are. Fragments would then have no adequate model and no PAR, and they would be judged on their measured results, every one of which meets the criterion. The judgement over the NOR, the range carried into the control strategy, which the PARs of MVM LRF set, and the design space do not change.

The selected model of step yield is intercept-only, with a mean of 98.2 % and a residual SD of 0.59 percentage points. The model with filtration volume lies 1.3 AICc units from it, with a coefficient of −0.34 points per coded unit that is not significant ($p = 0.173$). The residual plots of XMuLV LRF, fragments and step yield are given in Appendix C (Figure C.1, Figure C.2 and Figure C.3).

Figure 5.2 shows the measured results of both designs against filtration volume. The MVM LRF falls with volume at every pressure of the response-surface design, and at high volume the line of the model of all runs lies below the line with run 5 deleted. For XMuLV LRF and fragments no term was significant in either design.

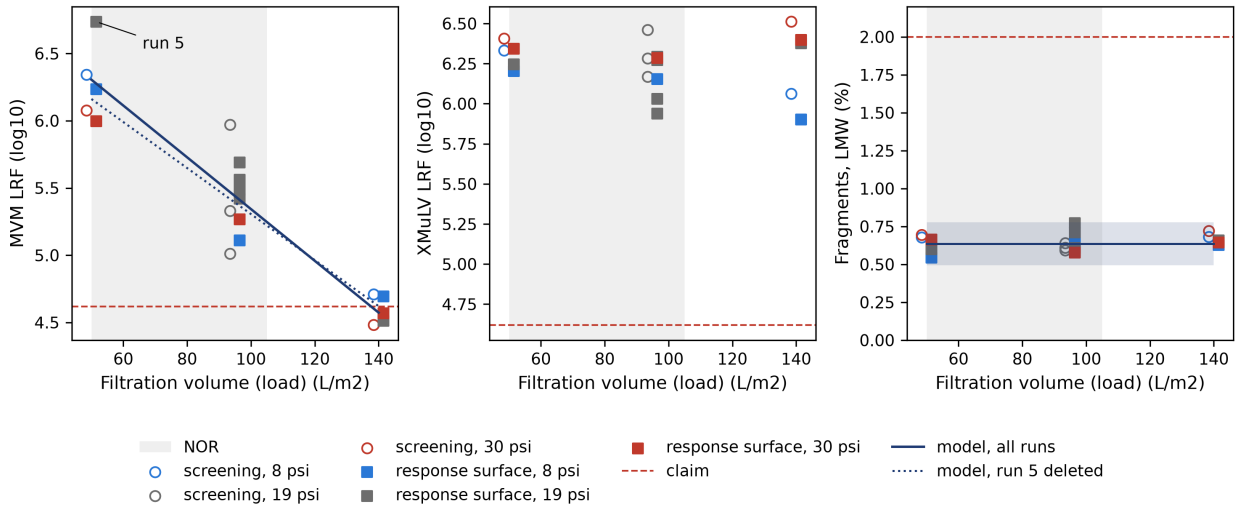


Figure 5.2: Measured results of both designs against filtration volume, with the claim or criterion of each output and the NOR shaded. Colour gives the filtration pressure, open circles the screening runs and filled squares the response-surface runs. For MVM LRF the lines are the selected model on all runs and with run 5 deleted. For fragments the line and band are the selected model and the 95 % prediction interval of a single run. XMuLV LRF has no adequate model and no line.

5.4 Mechanistic interpretation

The fall of MVM LRF with filtration volume is the effect that the mechanism of the step predicts. As the membrane fouls, the flux shifts toward the largest pores, through which the parvovirus can pass, and the shift grows with the volume filtered per unit area. The data show the effect as a fall of 1.73 log10 across the range studied, with a standard error of 0.19 log10.

The data do not show three other effects that PCP-009 expected from the same mechanism. First, the fall was expected to accelerate toward the top of the volume range. In the full quadratic of the planned runs the square of filtration volume is positive, 0.136 log10 per coded unit squared, which is the opposite sign, and it is not significant (p = 0.333). Within 50–140 L/m2 the decline is consistent with a straight line, and the mechanism as PCP-009 states it does not explain the absence of the acceleration. Second, MVM retention was expected to fall at high pressure, and at low pressure through diffusion. The main effect of pressure in the full quadratic is −0.034 log10 per coded unit (p = 0.706). The square of pressure is negative, −0.301, which would mean a lower LRF at both ends of the pressure range, the shape that the mechanism predicts, but it is not significant (p = 0.058) and was not selected. Third, the fall with volume was expected to be steeper at high pressure. The interaction in the full quadratic is 0.028 log10 per coded unit squared, with a standard error of 0.105 (p = 0.799), and in the screening design 0.02 log10 across the range (p = 0.965). Within the range studied no dependence of the volume effect on pressure was detected at the precision of these runs. A term that is not significant is not shown to be absent, and an effect of pressure of the size of these standard errors could exist within 8–30 psi.

For operation, the MVM LRF is therefore lowest at the highest volume filtered per square metre of membrane, and the claim is most at risk there. Within the pressures studied, pressure had no detected effect on any output, and the condition of uninterrupted filtration rests on the mechanism and not on these data, because the designs did not study an interruption of flow.

The results for XMuLV LRF agree with the expectation that a virus many times larger than the pores is retained independently of the operating parameters. No term was significant in either design, and every measured LRF met the claim. Every LRF in the record is a point value, and the record does not say whether any filtrate titre was below the limit of detection (Section 3.3), so the data do not show whether retention was complete.

Fragments did not depend on either parameter, and their level in the filtrate ranged from 0.54 to 0.77 %, against the criterion of 2.00 %. The expectation that the step does not change fragments cannot be tested on these data, because fragments were measured in the filtrate and not in the load. It rests on the size of the clipped species against the pores. Step yield did not depend on either parameter. The model with filtration volume, the next by AICc, falls with load as PCP-009 expected, but its coefficient is not significant (p = 0.173). The data therefore do not show whether the loss in the fouling layer grows faster than the share of the fixed hold-up falls (Section 2.1).

6 Design space

The design space of the step is the part of the characterized region in which every output with a criterion meets it at once, each on the statistic on which it is judged (PCMP-001 §6.2). A setting lies inside it when the mean predicted MVM LRF meets its claim and both ends of the 95 % prediction interval of a single run for fragments lie within their criterion. XMuLV LRF has no adequate model and draws no boundary, because every measured result met its claim. The region was evaluated on an evenly spaced grid of 11 by 11 points over the characterized region, 50–140 L/m2 by 8–30 psi.

Of the 121 grid points, 110 lie inside the design space, 91 % of the characterized region. The points outside are those at 140 L/m2, the top of the volume range, where the mean predicted MVM LRF falls below the claim at every pressure. Between the grid points the boundary lies at 137.6 L/m2 at every pressure, because the selected model has no pressure term. Fragments exclude no point. Drawn on the mean prediction of fragments as well, the design space is the same, because MVM LRF is judged on its mean in both and fragments exclude no point in either.

The NOR of both parameters lies inside the design space, and the design space lies inside the characterized region, which is the knowledge space of the study. Movement within the design space is not a change from the point of view of a regulatory filing under ICH Q8(R2), but it is evaluated under the change management system of the quality system (International Council for Harmonisation 2009, 2008). At the boundary itself a single run meets the claim only about half the time if the variability is symmetrical, because the boundary is drawn on the mean prediction, as PCMP-001 §6.2 sets it for viral clearance. At the least favourable NOR combination the lower end of the two-sided 95 % prediction interval of a single run, a one-sided 97.5 % limit, is 4.71 log10, above the claim. It is given for information.

Deleting run 5, the axial run at 50 L/m2 and 19 psi, or run 7, the axial run at 95 L/m2 and 8 psi, gives a selected model that meets the claim over the whole characterized region (Table 5.9). The design space is not confirmed at manufacturing scale at its edges, because virus cannot be spiked at that scale. The selected model of step yield is intercept-only, so the predicted mean yield is 98.2 % at every setting in the design space (PCP-009 §7).

7 Proven acceptable ranges

Each output of the step with a criterion is judged against its intermediate acceptance criterion (Table 7.1). MVM LRF and XMuLV LRF are each held to the platform claim of this step, 4.62 log10, the log reduction that the step delivered on the platform with X-Mab, Y-Mab and Z-Mab (PCMP-001 §7.4). The claims of the steps that clear each virus are summed and checked against the cumulative requirement of the drug substance. For XMuLV the claims of low-pH viral inactivation (6.60), anion exchange (5.50) and this step sum to 16.72 log10, against a requirement of 16.7 log10. For MVM the claims of anion exchange (4.00) and this step sum to 8.62 log10, against 8.6 log10. The LRF claimed for this step is therefore one of three claims for XMuLV and one of two for MVM. Each step is judged against its own claim, so a shortfall at this step is not made up by another step. Fragments carry the drug substance criterion, 2.00 %, because no later step changes them.

Table 7.1: Intermediate acceptance criteria for the virus filtration output and their derivation from the drug-substance criteria.

Attribute	DS criterion	Downstream worst case (platform or verified)		IAC	Basis
		Clearance factor	Increase		
Fragments, LMW (%)	≤ 2.00	1.00	0.00	≤ 2.00	DS criterion[a)]
XMuLV clearance (log10)	≥ 16.7[b)]	not applicable		≥ 4.62	Platform claim[c)]
MVM clearance (log10)	≥ 8.60[b)]	not applicable		≥ 4.62	Platform claim[c)]

[a)] No downstream step changes the attribute, so the drug-substance criterion applies.
[b)] Cumulative requirement across the steps that clear the virus.
[c)] Log reduction claimed for this step in the modular viral clearance claim.
DS, drug substance. IAC, intermediate acceptance criterion. Increase, in the unit of the criterion.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

The criteria of this step were derived without its data. The claims come from the platform history and the fragment criterion from the CQA register (PCMP-001 §7.3). The criteria in force equal their values derived in this way, because no verification changed them. No criterion of an earlier step rests on a platform worst case of this step, so this report has no worst case to verify. The fragment criteria of the earlier steps do assume that no later step changes fragments, and so that this step leaves its

load unchanged. That assumption rests on the mechanism of the step, because the record holds fragments in the filtrate and not in the load (Section 5.4).

Table 7.2 gives each output against its criterion over the NOR. MVM LRF met its claim, with a mean predicted LRF of 5.25 log10 at the least favourable NOR combination, the upper NOR edge of filtration volume. XMuLV LRF met its claim on its measured results. Fragments met their criterion, with an upper 95 % limit of a single run of 0.78 % within the NOR.

Table 7.2: Virus filtration outputs against their intermediate acceptance criteria.

Attribute	Criterion	Mean predicted by the response-surface model, or measured results where it has none[a)]		95 % limits of a single run over the NOR	Met across the NOR[b)]
		At the set-point	Range over the NOR		
MVM clearance (log10)	≥ 4.62	5.54	5.25 to 6.31	n.a.[c)]	Yes
XMuLV clearance (log10)	≥ 4.62	n.a.[a)]	5.90 to 6.51[a)]	n.a.[a)]	Yes
Fragments, LMW (%)	≤ 2.00	0.64	0.64 to 0.64	0.50 to 0.78	Yes

[a)] No adequate model (PCMP-001 §7.2): the range of the measured results of the designs, less the offset of the scale-down model where one is known, on which the output is judged.
[b)] Judged at every combination of NOR low, set-point and NOR high. Quality attributes: the 95 % prediction interval of a single run (t quantile, coefficient and residual uncertainty). Viral clearance: the mean prediction.
[c)] Judged on the mean prediction at the least favourable combination, the lower end of the range over the NOR.

Two analyses give the PAR of each parameter for each output with a criterion and an adequate model (Table 7.3). The first scans the parameter across its range studied on a grid of 201 points and gives the contiguous range over which the mean prediction of the response-surface model meets the criterion. The second is the NOR-propagated analysis of PCMP-001 §6.4. It scans the parameter across the same range on a grid of 81 points with the other parameter moving within its NOR. For viral clearance the other parameter is set at each grid point to its NOR low, its set-point and its NOR high, and the lowest of the three mean predicted LRF must meet the claim. For fragments, 2,000 values of the other parameter are drawn at each grid point from a normal distribution centred on its set-point, with an SD equal to the width of its NOR divided by 6, clipped to the range studied. Each draw carries the model residual and the uncertainty of the fitted coefficients, and the 97.5th percentile of the simulated batches must not exceed the criterion. The range reported is the contiguous range that contains the set-point, and the NOR-propagated range is the one carried into the control strategy (PCP-009 §8). An output with no adequate model has no PAR, and Table 7.3 marks it n.a. (PCMP-001 §7.2).

Table 7.3: Proven acceptable ranges of the parameters for each output with a criterion.

CQA	Parameter	Char. range	Unit	Criterion	PAR (set-point)	PAR (NOR)	Basis
MVM LRF (log_{10})	Filtration volume (load)	50–140	L/m2	≥ 4.62	50–137	50–136	selected model
	Filtration pressure	8–30	psi		8–30	8–30	
XMuLV LRF (log_{10})	Filtration volume (load)	50–140	L/m2	≥ 4.62	n.a.	n.a.	no adequate model
	Filtration pressure	8–30	psi		n.a.	n.a.	
Fragments (LMW, %, nrCE-SDS)	Filtration volume (load)	50–140	L/m2	≤ 2.00	50–140	50–140	selected model
	Filtration pressure	8–30	psi		8–30	8–30	

Edges rounded inward to three significant figures. Basis, selected model, the PAR comes from the selected response-surface model. No adequate model, the output has no PAR (n.a.), and every measured result met the claim.

The PAR of filtration volume for MVM ends inside the range studied in both analyses. It is 50–137 L/m2 in the first analysis and 50–136 L/m2 in the NOR-propagated analysis, as Table 7.3 rounds them. The mean prediction crosses the claim at 137.6 L/m2, which is the crossing value that PCP-009 §8 asks this report to state. Below that volume the selected model predicts a mean LRF above the claim at every pressure, because it has no pressure term. The NOR of filtration volume, 50–105 L/m2, lies inside its NOR-propagated PAR, which extends 31 L/m2 beyond the upper NOR edge. The lower edge of the PAR is the lower end of the range studied, and no run was made below it. Across the PAR the mean predicted LRF falls from 6.31 to 4.65 log10. The PAR of filtration pressure for MVM is the whole range studied in both analyses, and no claim is made beyond it.

Figure 7.1 shows the two analyses for MVM against filtration volume, with the claim drawn and the acceptable region shaded green. The two panels give nearly the same edge, because the selected model has no pressure term.

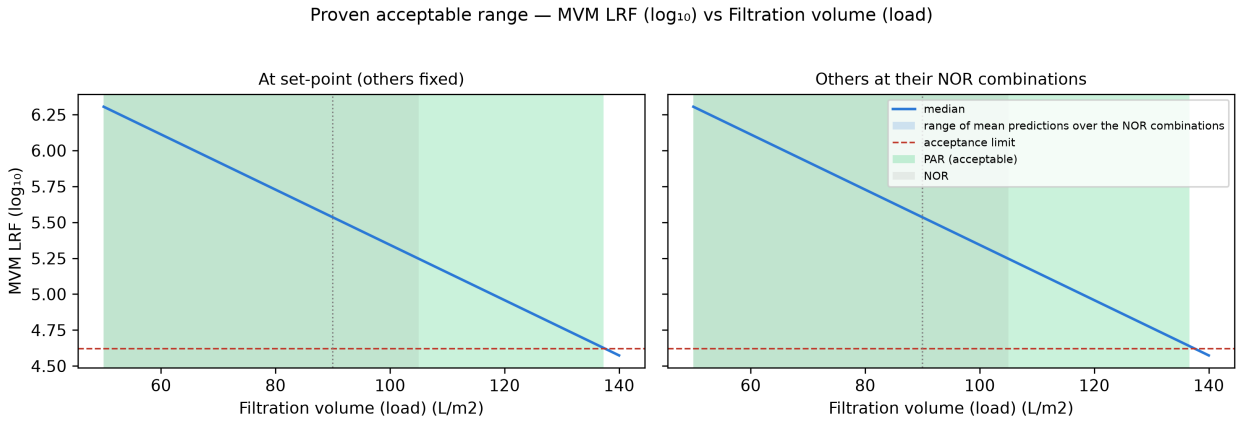


Figure 7.1: Proven acceptable range of filtration volume for MVM LRF, the first analysis (left) and the NOR-propagated analysis (right), with the claim of the step and the NOR.

XMuLV LRF has no adequate model and therefore no PAR (n.a. in Table 7.3). Every measured result met the claim (Section 5.3). The selected model of fragments is intercept-only, and its interval lies below the criterion everywhere, so both of their PARs are the whole range studied for both parameters. PCP-009 §8 planned a figure for each PAR, and only Figure 7.1 is given.

With two parameters and no interaction term in any selected model, the PARs and the design space describe one region. The design space is the characterized region below the volume at which MVM LRF crosses its claim, and the PARs are its edges. A movement of filtration volume within its PAR keeps the step inside the design space at any pressure within 8–30 psi, and a movement of pressure within its PAR does so at any volume within the PAR of volume.

8 Process capability and robustness

Capability at commercial scale was estimated from the integrated model of the train, which runs 2,000 simulated batches through every step with each process parameter drawn within its NOR (PCMP-001 §6.5). The cumulative viral clearance of a simulated batch is the sum of the clearances of the steps that clear the virus. For MVM these are anion exchange (PCR-008) and this step, and for XMuLV low-pH viral inactivation (PCR-006), anion exchange and this step. The capability of the cumulative clearance therefore depends on those steps as well as on this one.

Table 8.1: Capability of the drug substance attributes to which the step contributes, from the integrated model.

Attribute	Criterion	Mean ± SD	Cpk	Batches outside	Normal tail (per million)
Viral clearance — MVM (parvovirus)	≥ 8.60	10.38 ± 0.40	1.51	0 of 2,000	3.13
Viral clearance — XMuLV (enveloped)	≥ 16.7	19.88 ± 0.62	1.70	0 of 2,000	0.18
Fragments (LMW, clipped species)	≤ 2.00	0.64 ± 0.04	10.62	0 of 2,000	0.00

Viral clearance, the cumulative log10 reduction of the drug substance against the cumulative requirement. Normal tail, the tail probability of a normal distribution with the simulated mean and SD.

The Cpk of the cumulative MVM clearance is 1.51 and that of the cumulative XMuLV clearance 1.70, against the requirements of 8.6 and 16.7 log10 (Table 8.1). Both are above the target of 1.33 and the minimum of 1.0 of PCMP-001 §7.9, and MVM is the tighter of the two. The Cpk of fragments is 10.62 against the criterion of 2.00 %, also above the target. No simulated batch fell outside any of the three criteria. With none of 2,000 batches outside, the one-sided 95 % upper bound of the probability out of specification is 0.0015 for each attribute. That bound covers the sampling of the simulation and not the uncertainty of the integrated process model. The tail of a normal distribution with the simulated mean and SD gives 3.13 per million for MVM, 0.18 per million for XMuLV and a value that rounds to zero for fragments.

Within this step, the claim is judged on the response-surface model at the least favourable NOR combination (Section 7). The capability adds the variation of the parameters within their NOR and the clearance of the other steps. Fragments are judged per batch against their criterion and are tested at release of the drug substance.

9 Parameter classification

The parameters were classified under SOP-4001. Under the rule of PCP-009 §7 a parameter affects an attribute when a term that contains the parameter is significant at α = 0.05. The class of an affected parameter then follows from the size of its effect across its PAR and from the margin between the PAR and the NOR. Neither PCMP-001 nor SOP-4001 sets a number for either. The class of filtration volume below is therefore a judgement on the values given.

- Filtration volume (load), WC-CPP. A term containing the parameter is significant for MVM LRF ($p < 0.001$). Across its NOR-propagated PAR, 50–136 L/m2, the mean predicted MVM LRF falls by 1.65 log10. MVM clearance is an attribute of very high criticality. The PAR extends 31 L/m2 beyond the upper NOR edge, and its lower edge is the end of the range studied. The volumetric load is set by the volume of load applied to a filter of known area, so it can be held within its NOR. The parameter was classified as a WC-CPP on this margin and this control. With run 5 deleted the PAR would reach the end of the range studied, and the class would be the same.
- Filtration pressure, WC-CPP. No term containing the parameter is significant for any output, in either design (MVM LRF $p = 0.706$ for the main effect and $p = 0.058$ for the square in the full quadratic, XMuLV LRF $p = 0.053$ in its selected model, which is not adequate). No effect of pressure on an attribute was therefore detected within 8–30 psi, and the class does not follow from the rule of PCP-009 §7. The A-Mab case study classes both parameters of this step as WC-CPP, “since both are easy to control and have minimal impact on LRF when operated within their acceptable ranges” (CMC Biotech Working Group 2009, 157). The parameter was classified as a WC-CPP on that basis and on the scope of the viral clearance claim, which holds only for the pressures of the spiking runs, 8–30 psi, and for uninterrupted flow. A transient excursion to 22 psi in one run (DEV-009-01) lay inside the range studied. The PAR of the parameter is the whole range studied.

The step has two critical process parameters, both of the well-controlled kind. Neither parameter had a detected effect on step yield.

10 Contribution to the control strategy

The step contributes two parameter ranges, a condition of operation, a test of each filter after use and its share of the modular viral clearance claim. Filtration volume is held within its NOR of 50–105 L/m2 around the set-point of 90 L/m2, and an excursion is assessed against its NOR-propagated PAR of 50–136 L/m2. Filtration pressure is held within 10–20 psi around 13 psi, with a PAR of 8–30 psi. The range carried into the control strategy for each parameter is the intersection of its PARs over the outputs with an adequate model (PCMP-001 §7.6). For both parameters that is the PAR for MVM LRF, because the PARs for fragments cover the range studied. Filtration is uninterrupted, because the claim does not cover an interruption of flow. An interruption at commercial scale is assessed against the mechanism of the step and not against these ranges (PCP-009 §12).

Each filter is integrity tested after use under SOP-2012. The post-use test of a parvovirus filter is designed to detect a loss of small virus retention. A membrane that passes it has no defect large enough for the test to detect. The retention of MVM depends on that integrity at least as much as the retention of XMuLV does.

The step contributes 4.62 log10 to the cumulative MVM claim, beside anion exchange (4.00 log10, PCR-008). It contributes 4.62 log10 to the cumulative XMuLV claim, beside low-pH viral inactivation (6.60 log10, PCR-006) and anion exchange (5.50 log10, PCR-008). Viral clearance cannot be tested on a batch, so the claim rests on the operation of the step within its ranges. Fragments are tested at release of the drug substance by nrCE-SDS against the criterion of 2.00 %, and no later step changes them. PCMR-001 consolidates these ranges, the classes and the viral clearance claims with those of the other steps.

11 Discussion

As the volume filtered per square metre rises, the membrane fouls and the flow shifts to the larger pores, so the MVM LRF of the step falls. The study measured this fall as a straight line from 6.31 log10 at 50 L/m2 to 4.57 log10 at 140 L/m2. The claim of 4.62 log10 holds on the mean prediction up to 137.6 L/m2, 32.6 L/m2 beyond the upper NOR edge of 105 L/m2. The 3 measured MVM results below the claim were all at 140 L/m2, beyond that point. No effect of pressure was detected on any output, and XMuLV LRF, fragments and step yield did not depend on either parameter.

The scale-down model compared step yield alone with manufacturing scale, with an offset of −0.5 percentage points. No output with a criterion was compared at both scales. The clearance claims therefore rest on the spiking model, as ICH Q5A(R2) provides, and the fragment results on the mechanism of the step (International Council for Harmonisation 2023a). The flux decay profile, which PCP-009 named as the comparison that shows whether the model fouls like the commercial filter, is not recorded (PD-009-03). How far the fouling of the scale-down membrane represents the commercial filter is therefore not shown by these data, and the effect of filtration volume on MVM LRF is the effect at scale-down.

Deleting run 5 or run 7 carries the upper PAR edge of filtration volume to the end of the range studied (Table 5.9). The ranges reported come from all runs and are the narrower ones. Under every deletion the edge lies 30 L/m2 or more beyond the NOR, and no deletion changed the judgement over the NOR.

The data did not show the acceleration of the MVM decline at high load, the effect of pressure or the interaction between load and pressure that the mechanism led PCP-009 to expect (Section 5.4). The residual SD of a model sets the standard errors of its terms. For the selected model of MVM LRF it is 0.23 log10 (Table 5.5), and the slope of volume is 9.3 times its standard error. In the full quadratic of the planned runs the terms that PCP-009 expected have standard errors from 0.086 to 0.129 log10 in coded units, and none of them is significant (Table C.3). Effects of that size are not resolved by 12 runs. Interruptions of flow were not studied, so the condition of uninterrupted filtration rests on the mechanism.

The record lacks several results that PCP-009 required (Section 13.2). Two of them limit the claims. The record holds no spike titre and no limit of detection, so it cannot show whether any LRF is a lower bound. It also holds no integrity test result for any run, so it cannot show that each membrane was intact. The MVM results depend on that at least as much as the XMuLV results. The record names a membrane lot only in DEV-009-02, so variation between membrane lots was not studied.

The record does not name the runs of the two study events. The deletion of each response-surface run in turn covers an event that affected one of those runs. It does

not cover an event that affected several runs, such as the membrane lot of DEV-009-02 (Section 13.1).

12 Conclusions

1. Scale-down model (objective 1). The qualification compared step yield alone, with an offset of −0.5 percentage points (90 % confidence interval −1.2 to 0.3). Flux decay and the post-use integrity test are not recorded, so objective 1 was not met as PCP-009 states it. PCP-009 §7 requires this report to state what "the scale-down model is qualified" means for this step, given the outputs that the qualification compares. Only step yield was compared, and it has no criterion, so no conclusion was decided by an offset. The outputs with a criterion are judged on the scale-down results, the LRF on the spiking model and fragments on the mechanism of the step.
2. Effects (objective 2). Filtration volume lowers the mean MVM LRF by 1.73 log10 across 50–140 L/m2. No effect of filtration pressure was detected on any output, and no effect of either parameter on XMuLV LRF, fragments or step yield.
3. Models (objective 3). The selected model of MVM LRF passes the model F-test, shows no lack of fit and meets the limit on overfitting of PCMP-001 §7.2. It fails the criterion on residuals and influential runs on run 5, the axial run at 50 L/m2 and 19 psi. XMuLV LRF has no adequate model and is judged on its measured results. The selected models of fragments and step yield are intercept-only.
4. Proven acceptable ranges (objective 4). The NOR-propagated PAR of filtration volume is 50–136 L/m2 for MVM LRF and the whole range studied for fragments. The PAR of filtration pressure is the whole range studied, 8–30 psi, for both. Every PAR contains the NOR. XMuLV LRF has no adequate model and no PAR, and every measured result met its claim across the range studied.
5. Classification (objective 5). Both parameters are WC-CPP. The class of filtration pressure rests on the basis that the A-Mab case study gives and on the scope of the viral clearance claim, not on a detected effect.
6. Criteria. Each output with a criterion met its intermediate acceptance criterion across the NOR, MVM LRF at 5.25 log10 against 4.62, XMuLV LRF on measured results from 5.90 log10 against 4.62, and fragments at 0.78 % against 2.00 %.
7. Design space. The design space covers 91 % of the characterized region and excludes filtration volumes above 137.6 L/m2. Deleting run 5, the run at 50 L/m2 and 19 psi with the highest LRF at that volume, would carry the boundary to the end of the range studied.
8. Capability. The Cpk of the cumulative MVM clearance, 1.51, is the tightest of the attributes to which the step contributes, above the target of 1.33 and the minimum of 1.0. No simulated batch fell outside any criterion.
9. Success of the study. PCP-009 §7 asks for a qualified scale-down model, two viral clearance models accepted as predictive and a PAR containing the NOR for each parameter against each viral clearance response. The PARs of MVM LRF contain the NOR. The MVM model was used for prediction although it fails the criterion on residuals and influential runs (Section 13.2). The XMuLV model

was not accepted, so XMuLV LRF has no PAR, and its measured results met the claim across the whole range studied. The qualification compared step yield alone. The success criterion is therefore not met in full.

10. Record. Every run of the two study events was retained, as Version 1.0 recorded. The record does not name the runs, so the deletion of each run in turn covers only an event that affected a single response-surface run (Section 13.1). The results roll up into PCMR-001.

13 Study events and discrepancies from the plan

PCP-009 calls a departure from the plan a deviation. A process characterization study is not GMP manufacture and raises no deviation. This report records the events of execution that Version 1.0 recorded as study events, under the numbers Version 1.0 gave them, and every departure from PCP-009 as a discrepancy from the plan.

13.1 Study events

Version 1.0 recorded 2 study events at this step (Table 13.1). It recorded that every run concerned in either event was retained in the analysis. The record does not name the runs, so no run can be analysed at its recorded conditions or excluded alone. Every run of the response-surface design was therefore deleted in turn (Table 5.9), and no single deletion changed the judgement over the NOR. The deletion of each run in turn covers an event that affected one response-surface run. It does not cover an event that affected several runs. Nor does it cover the screening runs, whose results enter only the screening effects and the judgement of XMuLV LRF on its measured results, or the qualification runs, which enter only the offset of step yield.

Table 13.1: Study events of the step.

Event	Summary	Detected during	Outcome
DEV-009-01	Transient filtration-pressure excursion above NOR	in-process pressure monitoring	run retained in the analysis
DEV-009-02	Filter membrane lot flux below vendor-typical value	pre-use flux verification	run retained in the analysis

13.1.1 DEV-009-01, transient excursion of the filtration pressure

During in-process pressure monitoring the filtration pressure of one run rose for a short time to a maximum of 22 psi. The investigation attributed the excursion to an overshoot of the pump ramp. The record names neither the run, nor its set pressure, nor the duration of the excursion, so the run was analysed at its design setting. Version 1.0 recorded the peak as an excursion above the NOR, whose upper edge is 20

psi. The reference for an excursion is the set pressure of the run, because the designs set runs at 8, 19 and 30 psi. An overshoot rises above the set pressure, so the run was set below 22 psi, at 8 or 19 psi in a design or at the set-point of 13 psi in qualification. The peak lies inside the range studied, 8–30 psi, over which the designs estimated the effect of pressure, and no term containing pressure was significant for any output. If the run was a response-surface run, its deletion leaves the judgement over the NOR unchanged (Table 5.9). If it was a screening run, its result enters only the screening effects and the judgement of XMuLV LRF, every measured result of which met the claim. If it was a qualification run, it enters only the offset of step yield, on which no conclusion depends (Section 3.1). The run was retained in the analysis.

13.1.2 DEV-009-02, membrane lot flux below the value typical for the vendor

The pre-use flux verification of membrane lot LOT-VF-9331 (Meridian Membranes) gave a flux 12 % below the value typical for the vendor. The investigation attributed the low flux to variation of membrane permeability between lots. The record names the MVM infectivity method (AMV-3018) with the event, but it holds no result of an assessment by that method. Version 1.0 recorded that every run made on the lot was retained. The lot record gives the use of the lot as “verification run”, and no record names the runs. The lot may therefore have been used for several runs, and the deletion of one run at a time does not bound its effect. A lower permeability lowers the flux at a given pressure. Whether it changed the retention of MVM was not tested, so the effect of the event on the results cannot be assessed from the record. The record does not link the lot to run 5.

13.2 Discrepancies from the plan

Table 13.2 gives the discrepancies from PCP-009 that the register of discrepancies holds.

Table 13.2: Discrepancies from the plan PCP-009.

Discrepancy	Subject	Planned	Done	Reason
PD-009-01	Model selection for the response-surface outputs	PCP-009: the full quadratic model of each output (PCMP-001 §6 and §7.2)	for each output the scale chosen by the Box-Cox profile (the log scale where its 95 % interval contains 0 and excludes 1), the model with the lowest AICc among all hierarchical submodels of the full quadratic, a block term kept where the design has a second block, and that model judged against PCMP-001 §7.2; a scale-down offset taken on the scale of the model	the full quadratic carries terms the data do not support; selecting the model is standard practice, and INV-002 corrective action CA3 applies it to every report

Discrepancy	Subject	Planned	Done	Reason
PD-009-02	Conclusions resting on models that do not meet the criterion on residuals and influential runs	PCP-009: response-surface models meeting every model-adequacy criterion of PCMP-001 §7.2, used for the ranges and the design space, and each run with a Cook’s distance above 4/n examined by deleting it and repeating the judgement	the models of the outputs that do not meet the criterion on residual patterns and influential runs (a residual trend at the level of the residual tests, or a run with a Cook’s distance above the median of F(p, n − p)) used for the ranges and the design space, with the size of each trend stated against the criterion of its output and each run above the limit deleted in turn, the model selection repeated and the output judged over the NOR again	selected models of this step do not meet the criterion of PCMP-001 §7.2 on residual patterns and influential runs; whether a conclusion depends on the trend or on the run is shown by the size of the trend and by the deletion

Discrepancy	Subject	Planned	Done	Reason
PD-009-03	Scale-down qualification without flux decay and integrity results	PCP-009: the flux decay profile and the post-use integrity test compared at both scales, with load material, filtration temperature and hold-up volume per area matched (PCP-009 §5.1, Table 5.1)	step yield only; no flux decay or integrity result and no record of the matching; Version 1.0 asserted that every run passed its post-use test, with no result recorded	the record gives no reason
PD-009-04	Integrity tests and virus controls of the design runs not recorded	PCP-009: a pre-use and a post-use integrity test on every run; spike titres, limits of detection, lower-bound flags, cytotoxicity, interference and spike recovery controls; the achieved titre of each run; a clarified virus stock of small volume; and each sample identified by run and volumetric load (PCP-009 §5.2, §5.3, §5.4, §12)	none of these is in the record; the log reductions are recorded as results only	the record gives no reason

Discrepancy	Subject	Planned	Done	Reason
PD-009-05	Step yield and filtrate fractions not recorded as planned	PCP-009: step yield by protein mass balance from the ultraviolet absorbance of the load and the pooled filtrate of each spiked run (PCP-009 §5.2), an unspiked load and filtrate pool for yield (§5.4, which conflicts with §5.2), and filtrate fractions by volumetric load (§5.4)	the yield value of each spiked run, with no concentra- tions, volumes, assay or load sample taken before spiking; no filtrate fractions; the volume filtered in each run not recorded	the record gives no reason
PD-009-06	Load material and flux of the spiked runs not compared	PCP-009: the load material of each run identified, and the flux decay of the spiked runs compared with that of the qualification runs (PCP-009 §12)	neither; the record holds no flux data	the record gives no reason

Discrepancy	Subject	Planned	Done	Reason
PD-009-07	Analyses the plan does not contain	PCP-009: the analyses of PCP-009 §5.5, §7 and §8, with the examination of runs whose Cook's distance is above 4/n and one figure per PAR	the deletion of every response-surface run, a test of the curvature of the two designs against each other, the Shapiro-Wilk test and a single-run-limit reading of the Version 1.0 conditions, comparisons with near-equivalent and stepwise models, and a capability section; a PAR figure for filtration volume only	the plan delegates the reading of the Version 1.0 conditions to the report, and the influence analysis needs every run where the design is small

PD-009-01, the model selection (Section 3.5). The full quadratic that PCP-009 planned failed the limit on overfitting for every output with a criterion (Table 5.4). Under the plan no output with a criterion would therefore have had an adequate model. The verification of platform worst cases, from which some criteria of the campaign come, also uses the selected models. This report compared the criteria that the verification gives with the full quadratic models against the criteria in force. For this step they are the same, and so are the judgement of each output over the NOR and the PARs of MVM LRF.

PD-009-02, the MVM model used although it fails the criterion on residuals and influential runs. The PARs and the design space of MVM LRF come from the model of all runs. They are narrower than those with run 5 deleted, and no deletion changed the judgement over the NOR (Table 5.9).

PD-009-03, the scale-down qualification. No conclusion of this report depends on the qualification (Section 3.1). Without the flux decay profile, whether the scale-down membrane fouls like the commercial filter is not shown, so the effect of filtration volume on MVM LRF is the effect at scale-down.

PD-009-04, the integrity tests and virus controls of the design runs. The record cannot show that each membrane was intact, on which both LRF depend, or whether any LRF is a lower bound (Section 3.3). No run was excluded for a failed integrity test, because no test result is recorded.

PD-009-05, step yield and filtrate fractions. The record names no method for the step yield (Section 3.2). Without fractions the effect of filtration volume on MVM LRF is estimated between runs only (Section 3.4).

PD-009-06, the load material and the flux of the spiked runs. Whether the spiked load fouled the membrane differently from unspiked process material is not shown. PCP-009 §12 named that bias as largest at the top of the volume range, on which the claim depends.

PD-009-07, analyses the plan does not contain. They are the analyses that Section 3.5 lists as not in PCP-009. Of the PAR figures that the plan asked for, only Figure 7.1 is given.

Appendix A. Screening design matrix and responses

Table A.1 gives the settings and results of the 7 screening runs in natural units.

Table A.1: Virus filtration screening design, settings and results.

Run	Type	Factor settings		Responses			Fragments, LMW
		A	B	MVM LRF	XMuLV LRF	Yield	nrCE-SDS
		L/m2	psi	log10	log10	%	%
1	factorial	50	8	6.34	6.33	98.5	0.68
2		50	30	6.08	6.41	97.2	0.69
3		140	8	4.71	6.06	97.2	0.68
4		140	30	4.48	6.51	99.6	0.72
5	center	95	19	5.33	6.46	98.5	0.61
6		95	19	5.01	6.17	99.4	0.59
7		95	19	5.97	6.28	97.9	0.64

Factors: A, filtration volume (load). B, filtration pressure.
Not tested in this design: LMW by SEC.

Appendix B. Response-surface design matrix and responses

Table B.1 gives the settings and results of the 12 response-surface runs in natural units.

Table B.1: Virus filtration response-surface design, settings and results.

Run	Type	Factor settings		Responses			Fragments, LMW
		A	B	MVM LRF	XMuLV LRF	Yield	nrCE-SDS
		L/m2	psi	log10	log10	%	%
1	factorial	50	8	6.23	6.20	98.9	0.54
2		50	30	6.00	6.34	98.5	0.66
3		140	8	4.69	5.90	97.7	0.63
4		140	30	4.57	6.40	98.1	0.65
5	axial	50	19	6.74	6.25	98.1	0.60
6		140	19	4.51	6.38	97.9	0.66
7		95	8	5.11	6.15	97.5	0.64
8		95	30	5.27	6.28	98.5	0.58
9	center	95	19	5.48	5.94	99.0	0.77
10		95	19	5.56	6.29	98.9	0.59
11		95	19	5.69	6.03	97.2	0.71
12		95	19	5.42	6.27	98.2	0.61

Factors: A, filtration volume (load). B, filtration pressure.
Not tested in this design: LMW by SEC.

Appendix C. Full effect and coefficient tables

Table C.1 gives every effect of the screening models with its coefficient, Table C.2 the coefficients of every selected model of the response-surface design, and Table C.3 the coefficients of the full quadratic models that PCP-009 planned, which Version 1.0 used.

Table C.1: Effects and coefficients of the screening models.

Response	Term	Effect	Coefficient	Std. err.	t	p-value
MVM LRF (log10)	A	−1.614	−0.807	0.200	−4.04	0.027
	B	−0.246	−0.123	0.200	−0.62	0.581
	AB	0.019	0.010	0.200	0.05	0.965
XMuLV LRF (log10)	A	−0.082	−0.041	0.061	−0.67	0.548
	B	0.262	0.131	0.061	2.16	0.120
	AB	0.188	0.094	0.061	1.54	0.220
Fragments, LMW (% points)	A	0.015	0.007	0.032	0.23	0.832
	B	0.028	0.014	0.032	0.44	0.687
	AB	0.011	0.006	0.032	0.18	0.869
Step yield (% points)	A	0.567	0.283	0.365	0.78	0.494
	B	0.539	0.269	0.365	0.74	0.514
	AB	1.811	0.905	0.365	2.48	0.089

A, filtration volume. B, filtration pressure. AB, their interaction. Coefficients in coded units. Std. err., the standard error of the coefficient.

Table C.2: Coefficients of the selected models of the response-surface design.

Output	Term	Coefficient	Std. err.	t	p-value
MVM LRF (log10)	Intercept	5.439	0.066	82.44	< 0.001
	A	−0.866	0.093	−9.28	< 0.001
XMuLV LRF (log10)	Intercept	6.203	0.041	150.51	< 0.001
	B	0.128	0.058	2.19	0.053
Fragments, LMW (%)	Intercept	0.636	0.018	35.82	< 0.001
Step yield (%)	Intercept	98.208	0.170	578.40	< 0.001

Coefficients in coded units. The selected model of XMuLV LRF is not adequate and sets no range.

Table C.3: Coefficients of the full quadratic models on the planned runs, the model of PCP-009 and of Version 1.0.

Output	Term	Coefficient	Std. err.	p-value
MVM LRF (log10)	Intercept	5.522	0.096	< 0.001
	A	−0.866	0.086	< 0.001
	B	−0.034	0.086	0.706
	AB	0.028	0.105	0.799
	A^2	0.136	0.129	0.333
	B^2	−0.301	0.129	0.058
XMuLV LRF (log10)	Intercept	6.165	0.072	< 0.001
	A	−0.019	0.064	0.776
	B	0.128	0.064	0.094
	AB	0.090	0.079	0.299
	A^2	0.085	0.096	0.413
	B^2	−0.007	0.096	0.944
Fragments, LMW (%)	Intercept	0.661	0.033	< 0.001
	A	0.020	0.029	0.516
	B	0.012	0.029	0.696
	AB	−0.025	0.036	0.504
	A^2	−0.016	0.044	0.733
	B^2	−0.034	0.044	0.470
Step yield (%)	Intercept	98.211	0.313	< 0.001
	A	−0.336	0.280	0.275
	B	0.160	0.280	0.587
	AB	0.199	0.342	0.583
	A^2	−0.029	0.419	0.946
	B^2	0.023	0.419	0.958

Coefficients in coded units. A^2, B^2, the squares. These models do not meet the limit on overfitting and are not used for any conclusion of Version 2.0.

Figure C.1, Figure C.2 and Figure C.3 give the residual plots of the selected models of the other three outputs, which PCP-009 §5.5 asks for every response. An intercept-only model predicts one value, so its residuals lie on one vertical line against the prediction, and its predictions on one horizontal line against the measured results.

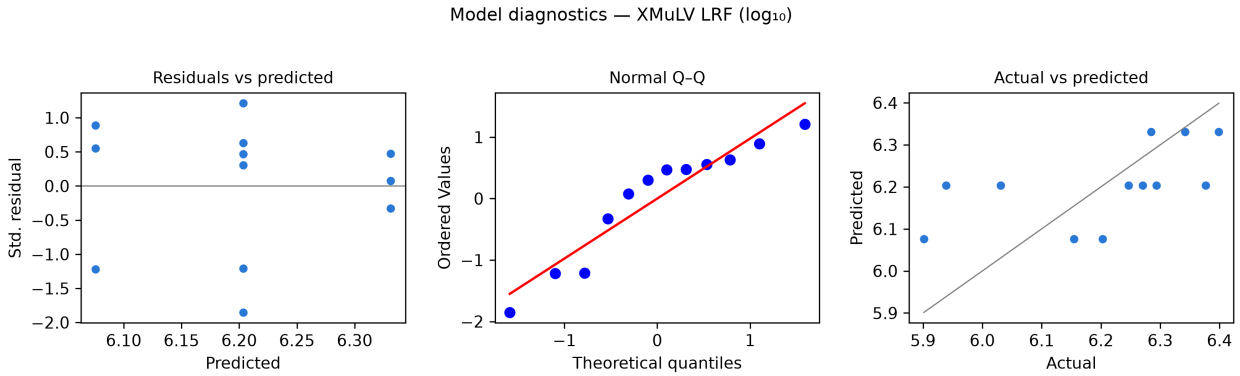


Figure C.1: Diagnostics of the selected model of XMuLV LRF, which is not adequate and sets no range.

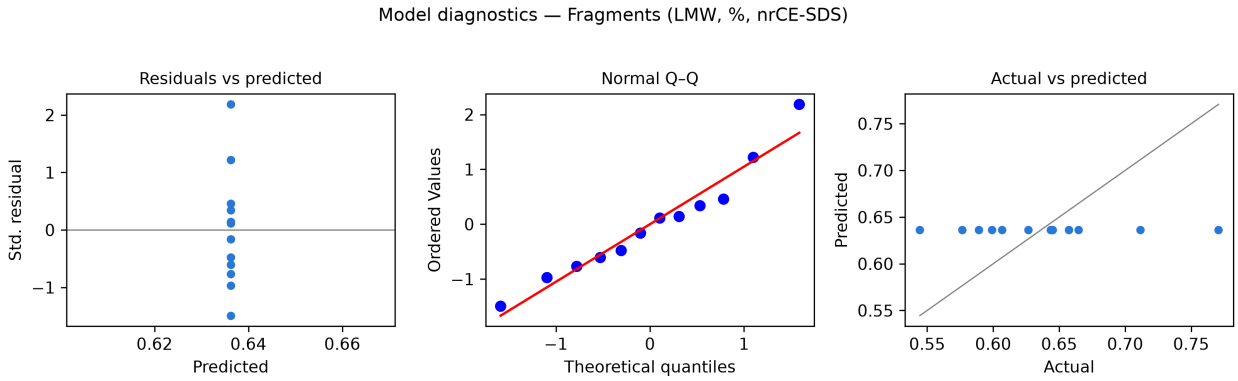


Figure C.2: Diagnostics of the selected model of fragments, an intercept-only model.

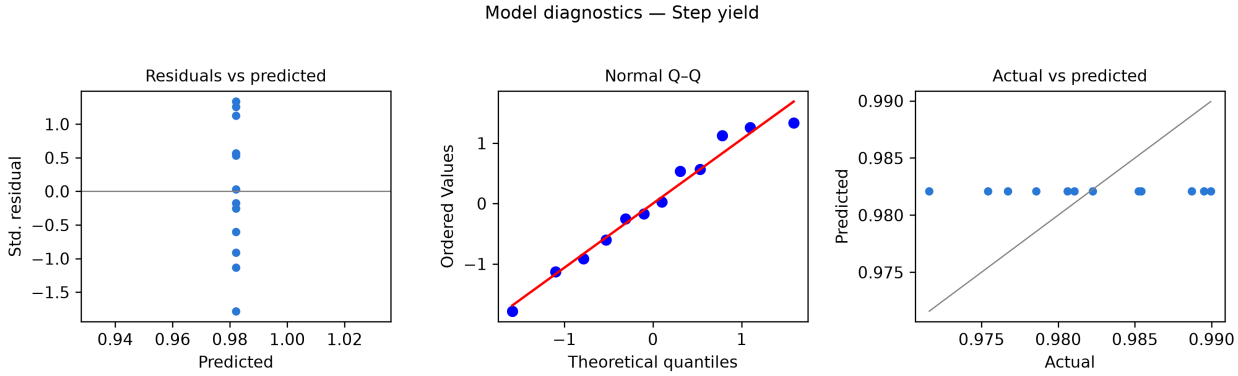


Figure C.3: Diagnostics of the selected model of step yield, an intercept-only model, with the yield as a fraction.

Appendix D. Analytical methods summary

Table D.1 summarises the methods behind the outputs of the designs.

Table D.1: Analytical methods for the outputs of the designs.

Output	Technique	AMV	Reportable value	LOQ
MVM LRF	Infectivity titre by TCID50, with qPCR confirmation	AMV-3018	LRF, log10	1.40 log10 TCID50/mL
XMuLV LRF	Infectivity titre by TCID50	AMV-3017	LRF, log10	1.50 log10 TCID50/mL
Fragments, LMW	Purity by nrCE-SDS	AMV-3020	LMW, % of total	0.1 %
Step yield	not stated in the record	–	%	–

AMV, the method validation, whose title the table of procedures and method validations of the analytical methods section gives. LOQ, the limit of quantitation of the titre or of the attribute, as the method record gives it. The LRF is calculated from the titres of the spiked load and of the filtrate.

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